

Respiratory practice booklet for first-year nursing students

Based on Ellie's bronchiolitis scenario, but written in simpler placement-friendly language. This guide keeps the core anatomy, bronchiolitis, feeding, development, family-centred care, and MDT content, then adds the other common respiratory conditions you are likely to compare on placement.

Ellie

one child threaded through the guide

4

common respiratory conditions

A–E

placement-safe structure

Family

psychosocial care included

1 Meet the child before you chase the theory

Ellie is the thread running through the booklet. If you can explain the theory in a way that makes sense for Ellie, your answer will sound more clinical and more student-nurse friendly.

SCENARIO

Ellie is **10 months old**. She has had **3 days of worsening cough, runny nose, and difficulty feeding**. Her respiratory rate is **62 breaths/min**, her **SpO₂ is 90% on air**, and she has **intercostal and subcostal recession** with **grunting**. She has also lost weight since her last GP visit. Ellie is admitted with **severe bronchiolitis**. Her mum Sara is a single parent with two older children at home and is visibly distressed.

RR 62/min

SpO₂ 90% on air

Recession

Grunting

Poor feeding

Weight loss

WHAT MATTERS MOST

Ellie is not just "wheezy". The big concerns are **work of breathing, oxygenation, feeding, fatigue**, and how quickly a young infant can deteriorate.

HOW TO ANSWER WELL

Keep linking the science back to the child. Say what is happening, why it matters, and what you would do or escalate.

WHAT THIS BOOKLET COVERS

Anatomy, breathing mechanics, bronchiolitis, signs, feeding, development, family-centred care, MDT support, and other common respiratory conditions.

SECTION	WHAT TO REMEMBER
Normal respiratory system	Air moves from upper airway to alveoli, where gas exchange happens across a thin membrane.
Breathing mechanics	Inspiration increases thoracic volume; expiration is usually passive but can become active in distress.
Bronchiolitis	Inflammation, oedema, and mucus narrow the bronchioles, raising resistance and reducing ventilation.
Child and family impact	Infants rely on their caregivers. Distress, separation, and practical family pressures all matter clinically.

2

The respiratory system and how air moves

First-year students do not need overcomplicated wording here. Keep it simple: where the air goes, where gas exchange happens, and how pressure changes move air in and out.

NORMAL STRUCTURE AND FUNCTION

Air enters through the **nose**, then passes through the **pharynx, larynx, trachea, bronchi, and bronchioles** before reaching the **alveoli**. The alveoli are where gas exchange happens. They work well because they have a **large surface area**, a **thin membrane**, and a **rich capillary network**.

Large surface area

Thin membrane

Good blood supply

WHY FICK'S LAW MATTERS

You do not need a complex formula in every answer. The key idea is that gas exchange is easier when the surface area is big, the membrane is thin, and the pressure gradient is good. In bronchiolitis, mucus and inflammation make this process less efficient.

INSPIRATION

During inspiration, the **diaphragm contracts and moves down**. The **external intercostal muscles** help lift the ribs up and out. This makes the thoracic cavity bigger, so intrapulmonary pressure falls and air moves into the lungs.

EXPIRATION

Expiration is usually **passive** because the lungs recoil as the diaphragm and intercostals relax. But when a child is struggling, expiration can become **active**, with extra muscle effort. Ellie's grunting tells you normal passive breathing is no longer enough.

BOYLE'S LAW IN PLAIN ENGLISH

When volume goes up, pressure goes down. That is why air moves in during inspiration. **When volume goes down, pressure goes up.** That is why air moves out during expiration. This is the simplest way to explain normal breathing in an exam answer.

3

Bronchiolitis, signs, and feeding

This is the core clinical section: what bronchiolitis is, why infants struggle, what the signs mean, and why feeding becomes such a big issue.

WHAT BRONCHIOLITIS IS

Bronchiolitis is a **viral lower respiratory tract infection**, often caused by **RSV**. The bronchioles become **inflamed, swollen, and full of mucus**. Because bronchioles are already tiny, even a small amount of swelling can reduce airflow a lot.

WHY AIRFLOW GETS WORSE

Narrower bronchioles mean **higher airway resistance**. Less air reaches the alveoli, so ventilation falls while blood flow continues. This creates a **low V/Q mismatch**, and oxygen saturation can drop.

SIGN	WHAT IT MEANS	WHY NURSES CARE
Tachypnoea	The child is trying to keep ventilation going.	Repeat observations and watch for fatigue.
Recession	The child is using more effort to breathe in.	Shows increased work of breathing, not a minor sign.
Grunting	The child is trying to keep airways and alveoli open.	Red flag. Escalate and reassess urgently.
Poor feeding	Breathing and feeding are competing for energy.	Hydration, nutrition, and NG support may need discussion.
Low sats / lethargy	Hypoxaemia and tiring.	Can signal worsening deterioration.

FEEDING AND BRONCHIOLITIS

Infants need to **suck, swallow, and breathe in a coordinated way**. When breathing is already hard work, feeding becomes harder too. That is why Ellie's **poor feeding and weight loss** are clinically important, not just "extra symptoms".

SIMPLE EXAM LINE

"Ellie's poor feeding matters because infants cannot feed well when respiratory distress is high, and this increases the risk of dehydration, fatigue, and further deterioration."

4

How hospitalisation affects Ellie and Sara

Your source document was strong on this, so I've kept it in. First-year students often forget that good paediatric care is not just physiology.

ERIKSON

Ellie is in the stage of **trust versus mistrust**. Consistent care and comfort help infants feel safe. Separation and inconsistent caregiving can increase distress.

BOWLBY

Ellie is strongly attached to her mum. Separation can cause distress and insecurity, especially in an unfamiliar ward environment.

PIAGET

Ellie is in the **sensorimotor stage**. She may not understand that her mum will leave and come back, which can increase anxiety.

WHY DISTRESS MATTERS PHYSICALLY

Crying raises **heart rate, oxygen demand, and work of breathing**. In a child already struggling to breathe, emotional distress can make the respiratory picture worse.

SARA'S PRACTICAL PRESSURES

Sara is balancing Ellie's admission with **two older children, travel, money worries, and emotional stress**. Those pressures affect how well the family can cope.

IMPACT	WHAT IT MIGHT LOOK LIKE	NURSING RESPONSE
Separation distress	Ellie becomes more upset when Sara leaves or staff are unfamiliar.	Promote parental presence and consistent caregiving where possible.
Parental anxiety	Sara looks overwhelmed, worried, guilty, or unable to absorb information.	Use clear explanations, active listening, and repeat important advice calmly.
Isolation and stress	The family feels unsupported or stretched too thin.	Offer practical signposting and involve the wider team early.

5

Who supports Ellie during admission and after discharge?

This section keeps your MDT ideas but turns them into simpler placement notes. Think about who does what, not just naming job titles.

DURING ADMISSION

- **Staff nurse:** observations, oxygen, feeding support, reassurance, escalation.
- **Doctor / paediatric team:** diagnosis, review, and treatment planning.
- **CNS or specialist support:** extra respiratory or feeding advice where available.
- **Play specialist:** helps reduce distress and support development.

AFTER DISCHARGE

- **GP:** recovery review, ongoing feeding and weight checks.
- **Health visitor:** development support, home reassurance, and monitoring.
- **Family / carers:** watching breathing, feeding, fluids, and seeking help if worried.

MARTHA'S RULE

Martha's Rule is about **listening to concerns about deterioration and acting on them**. Families and carers should know that if they are worried a child is getting worse, those worries matter and should be taken seriously.

GOOD FAMILY-CENTRED LINE

"Parents often know when something is not right before the numbers change, so their concerns are an important part of assessment."

Observe

Keep reassessing breathing, sats, heart rate, feeding, and alertness.

Explain

Tell parents what you are watching and why the signs matter.

Support

Think oxygenation, fluids, feeding support, and emotional reassurance.

Safety-net

Make sure families know what worsening signs to act on after discharge.

6 What else could a breathless child have?

This is the extra comparison page you asked for. It is designed for first-year pattern recognition, not specialist diagnosis.

CONDITION	TYPICAL PATTERN	SOUND / AIRWAY CLUE	WHAT NURSES SHOULD NOTICE	FIRST SAFE THOUGHTS
Bronchiolitis	Usually infants, starts like a cold, then breathing and feeding worsen	Wheeze and/or crackles, lower small-airway problem	Poor feeding, recession, fatigue, low sats	Supportive care, oxygenation, hydration, feeding support, escalation if tiring
Croup	Barking cough, hoarse voice, often worse at night	Stridor , think upper airway narrowing	Is the child calm or distressed? Is stridor present at rest?	Keep the child calm, avoid upsetting them, get urgent review if more severe
Viral wheeze / asthma	Wheezy child, often previous similar episodes or triggers	Expiratory wheeze, lower airway bronchospasm pattern	How breathless are they? Do they respond to inhaler/spacer?	Think local wheeze pathway, treatment response, and close reassessment
Pneumonia	Fever, cough, increased work of breathing, child may look generally more unwell	Sometimes crackles, sometimes reduced air entry	High fever, focal chest signs, lethargy, hydration status	Think infection, fluids, sepsis awareness, and escalation if the child looks very unwell

EASY COMPARISON

Stridor = think upper airway.

Wheeze = think lower airway.

Poor feeding in an infant = take it seriously.

High fever plus focal crackles = think pneumonia more strongly.

ESCALATE BEFORE YOU OVER-LABEL

If a child is blue, apnoeic, very sleepy, drooling, or struggling to maintain oxygenation, your priority is airway, breathing, and urgent review, not arguing over the exact diagnosis.

7 What to say on placement

Students often know the signs but get stuck on the wording. Calm, simple language usually sounds better than overcomplicated theory.

SIMPLE SBAR

Situation: "I'm concerned about Ellie because her breathing looks harder work."

Background: "She has had 3 days of cough, coryza, and reduced feeding."

Assessment: "Her respiratory rate is high, her sats are low, and she has recession with grunting."

Recommendation: "Please review urgently while we continue observations and supportive care."

PARENT-FRIENDLY WORDING

"Your child is working quite hard to breathe, so we're watching closely."

"The important things for us are breathing, oxygen levels, and whether Ellie can drink enough."

"If she becomes more sleepy, more blue, or breathing gets harder, tell us straight away."

FIRST-YEAR REMINDERS

- Describe what you can see before trying to sound advanced.
- Link signs to what they mean and what you would do.
- Feeding matters in infants.
- Escalating early is safe practice, not failure.

SOURCES CHECKED FOR THIS VERSION

NICE bronchiolitis guidance, NICE pneumonia guidance, NHS croup, NHS asthma, and NHS England Martha's Rule. This booklet is a simplified student resource, so treatment detail is intentionally high-level.

ONE-LINE SUMMARY

When a child is breathless, first-year nurses should focus on work of breathing, oxygenation, feeding, and family concern, then escalate early and communicate clearly.